

Day 3 — Assignment

Seaborn Visualization — abalone.csv

Work through these tasks using only the Seaborn functions covered today. Target time: 30–45 minutes. We're switching back to `abalone.csv` from Day 1 — same data, brand-new way of visualizing it.

Section 1 — Relational & Categorical Plots

Task 1: Import `seaborn` and load `abalone.csv`. Create a lineplot with `Length` on the x-axis and `Diameter` on the y-axis.

Task 2: Create a scatterplot with `Whole weight` on the x-axis and `Shell weight` on the y-axis.

Task 3: Create a barplot showing the mean number of `Rings` for each `Sex` category.

Task 4: Create a countplot showing how many rows exist for each `Sex` category.

Task 5: Create a boxplot comparing `Diameter` across the three `Sex` categories.

Section 2 — Distribution Plots

Task 6: Create a histplot of the `Height` column.

Task 7: Create a pairplot using the `Length`, `Diameter`, `Height`, and `Rings` columns, with `hue` set to `Sex`.

Section 3 — Multi-Plot Layouts & Styling

Task 8: Using `plt.subplot()`, place a scatterplot of `Length` vs `Whole weight` side by side with a boxplot of `Whole weight` by `Sex`, in one figure.

Task 9: Use `sns.FacetGrid()` to create one panel per `Sex` category, each showing a histogram of `Rings`.

Task 10: Set the Seaborn style to `'whitegrid'`, then create a lineplot of `Length` vs `Height` and remove the top/right border with `sns.despine()`.